

2024 Year in Review

Global Internet Traffic · Technology Adoption · Security Threats

A data-driven analysis of Internet patterns observed through Cloudflare's global network during 2024. Covering traffic growth, popular services, technology adoption, connectivity disruptions, and the evolving security landscape.

17%

Traffic Growth

225

Disruptions

6.5%

Traffic Mitigated

17% Global Traffic Growth and Starlink's 3.3x Surge

Continued upward trend driven by Internet reliance across communication, commerce, entertainment, and connectivity

17%

Global Internet traffic growth YoY in 2024

3.3x

Starlink traffic surge — satellite connectivity to underserved regions

41%

Global traffic from mobile devices in 2024

Traffic trend starts mid-January to normalize for post-holiday activity. Starlink trend is scaled relative to peak observed request volume.



Google Leads Top 10 Services as Generative AI Rises

Rankings based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver · Google tops ranking for second consecutive year

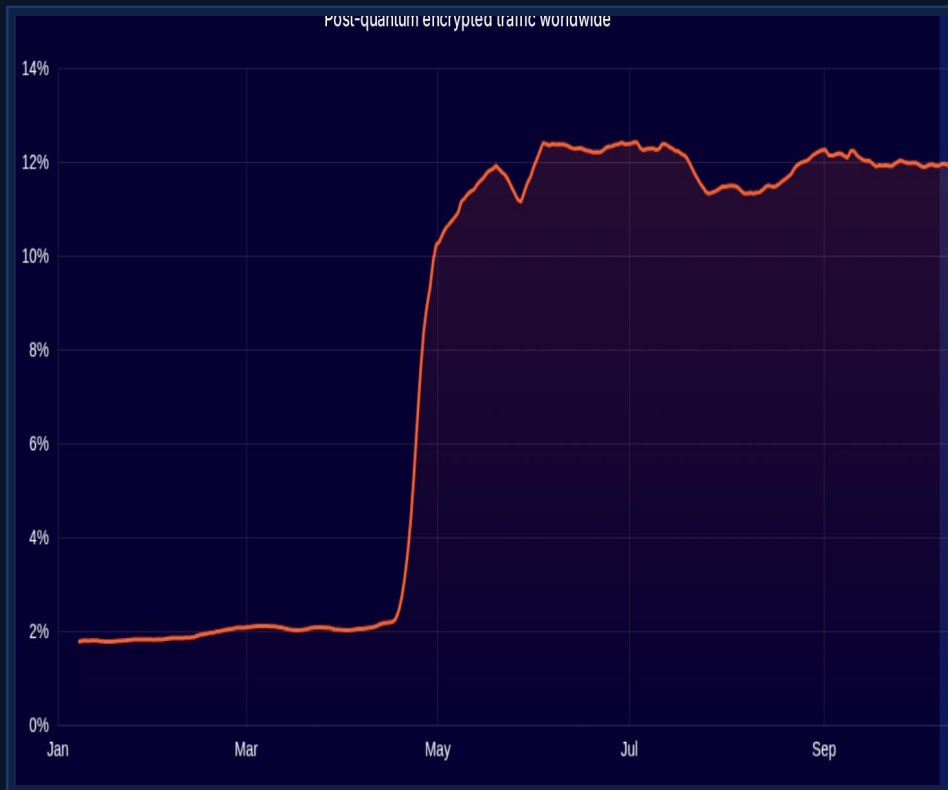
RANK	SERVICE
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Rank	Service
#1	Google
#2	Facebook
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#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI continued its rapid rise throughout 2024 — while the Metaverse receded from relevance.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Technology adoption shifts reshaping security, protocol usage, and developer preferences across the Internet



2% → 13%

Post-quantum TLS 1.3 adoption after Chrome 124 (Apr 17)

21%

HTTP/3 adoption — faster, encrypted by default over QUIC

~12%

Go leads API client languages — most popular for automated requests

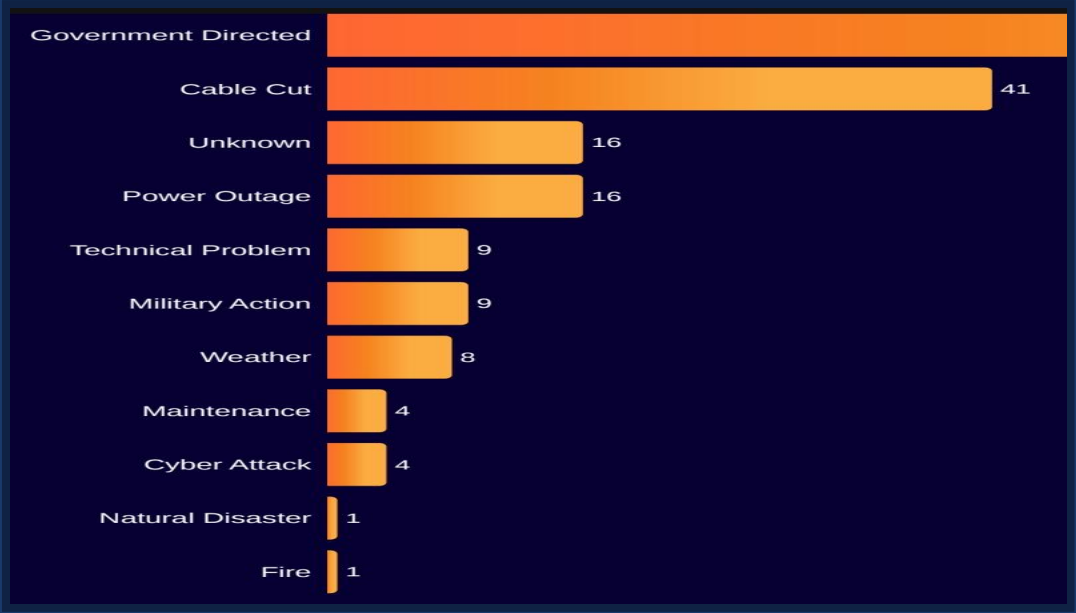
Metric

Value

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare Radar Outage Center tracked 225 major Internet disruptions globally in 2024 — scope and duration measured via Cloudflare traffic data

CAUSE	COUNT
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1



Government-directed shutdowns (116+) account for over half of all disruptions — primarily exam-related or political censorship.

Cable cuts (41 incidents) remain a significant physical infrastructure risk to Internet reliability.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security findings from Cloudflare's global network — DDoS, bot traffic, email threats, and vulnerability exploitation in 2024

6.5%

Of all traffic processed was mitigated — indicating pervasive, ongoing threats at scale

68%

Of global bot traffic originated from just the top 10 countries — highly concentrated sources

4.3%

Of processed emails were malicious; 43% of those contained deceptive links to trick recipients

Log4j

CVE-2021-44228 from Dec 2021 is still actively exploited 3+ years later — persistence of known vulns

MOST ATTACKED SECTOR

Gambling & Games

RPKI Valid Routes: +6.4% increase in 2024 — improving Internet routing security. | **.bar TLD:** Largest source of malicious and spam email observed.

Five Takeaways Shaping the 2025 Internet Outlook

Strategic findings from Cloudflare Radar 2024 to inform planning for network and security leaders

01

Traffic Growth is Structural

17% YoY growth with Starlink's 3.3x surge reflects durable demand — plan capacity and CDN strategy for accelerating baselines, especially in emerging and satellite-connected markets.

02

Post-Quantum Encryption is Now

The 2%→13% TLS 1.3 jump driven by Chrome 124 shows browser vendors can reshape cryptographic baselines overnight. Evaluate quantum-resistant readiness across your entire TLS infrastructure.

03

Connectivity is a Geopolitical Variable

Government-directed shutdowns caused 116+ of 225 disruptions in 2024 — more than all other causes combined. Multi-path routing and resilience planning are essential for global operations.

04

Known Vulnerabilities Remain Live Threats

Log4j (2021) is still being actively exploited. 6.5% traffic mitigation and Gambling/Games as top target sector underscore the need for continuous patching and DDoS-resilient architectures.

05

Developer Ecosystem is Shifting

Go leading API client languages at ~12% reflects a shift toward performance-first, compiled languages in backend automation. Factor this into API gateway and observability tooling investments.